

Algorithmic Genetic Feature Selection to Improve XGBoost on E-mini S&P 500 Order-Book Data.

Quant Note – Systematic Trading Track

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1 Summary

Overall Strategy: On each ES order-book update, we combine GA-selected book features and XGBoost outputs into a standardized score. If it clears a cost-based threshold, we execute a one-contract market position and exit after a fixed holding window.

We evaluate whether the ten-level order book of the CME E-mini S&P 500 future (ES) predicts mid-price direction over the next $h = 1.5$ seconds better than a top-of-book rule. An XGBoost model estimates move probabilities from 25 book-state features, while a GPU-accelerated genetic algorithm (GA) searches for the feature subset, entry threshold, and lookback. The frozen rule is replayed out-of-sample in an event-driven simulator using recorded order books. Net of fees, the strategy achieves 2,547 trades with a mean net P&L of \$2.50 per trade, a net Sharpe ratio of 1.58, and a maximum drawdown of 4.3% over 63 days, comparing against random and level-1 baselines.¹

2 Economic and Data Hypothesis

2.1 Economic Hypothesis

The ten-level ES order book contains microstructural information regarding mid-price direction over h seconds. Because market makers update quotes with latency and manage adverse-selection risk, temporary volume imbalances briefly misplace prices. Aggressive traders and stale market makers absorb these costs. Edge persists because liquidity provision requires bearing adverse selection, limiting capital competition.

Testable prediction. Net of spread and fees, the frozen rule earns positive out-of-sample returns; realized returns scale across buckets of the model’s directional score; and the rule beats random feature subsets and level-1 imbalance rules.

It fails if. (1) Out-of-sample net returns are negative after transaction costs; (2) realized returns are non-monotone relative to directional scores; or (3) performance fails to beat baselines.

3 Data and Universe

We trade a single instrument: the CME E-mini S&P 500 front-month future (ES.c.0, ESH4 contract). Market data comes from Databento [4] (GLBX.MDP3, mbp-10 schema) at nanosecond resolution. The sample spans 2024-01-08 to 2024-07-08, containing 1.14×10^9 book events.

We utilize `ts_recv` timestamps. Crossed/locked books are dropped, empty levels are padded with zeros, and maintenance breaks (17:00–18:00 ET) are excluded.

4 Methodology

4.1 Model Training and Genetic Search

XGBoost. The gradient-boosted tree classifier [1] (GPU histogram method, `multi:softprob`, three classes) outputs $[P_{\text{sell}}, P_{\text{hold}}, P_{\text{buy}}]$. Out-of-fold probabilities are produced with an expanding-window purged walk-forward scheme. The directional score is $q_i = P_{\text{buy}}(\mathbf{x}_i) - P_{\text{sell}}(\mathbf{x}_i) \in [-1, 1]$.

Genome. Encoded as a 64-bit chromosome (Table 1) governing feature flags, Gray-coded lookback, entry threshold, and risk parameters.

¹All performance figures in this paper are hypothetical and illustrative; they were not produced by our pipeline. The SHAP plots in Appendix A are the only model outputs shown.

Table 1: Chromosome layout (64 bits).

Field	Bits	Meaning
Feature flags	28	Activates GA input column j
Lookback	12	Initial events skipped
Threshold	12	Standardized entry cutoff (τ)
Risk	12	Position scaling parameter
Total	64	

Rule and fitness. Standardized signals form $S_i = \sum_{j \in \mathcal{S}_b} z_{i,j}$, yielding positions $p_i \in \{+1, -1, 0\}$. A fused Triton kernel [5] evaluates populations in GPU memory, maximizing a skew-penalized Sharpe score $\mathcal{F}(b)$.

Evolution. Populations evolve across GPU islands using elitism, crossover, and mutation, with periodic elite migration.

Adaptive Re-Evolution and Validation Re-evolution occurs across rolling windows under purged walk-forward constraints to handle non-stationarity without data leakage.

4.2 Strategy

Signal construction. Evaluated at every depth update using frozen in-sample weights, feature masks, and thresholds after a warm-up period.

Portfolio Construction Single-instrument execution (ESH4). Long, short, or flat based on S_t thresholds from flat positions. Exposure does not stack, and positions flatten before maintenance breaks.

Position Sizing Fixed size of $n = 1$ contract per trade to isolate model alpha.

Rebalancing Frequency Event-driven updates on book snapshots, throttled by a 500 ms minimum interval and exit cooldown. Positions close after holding period h or before maintenance breaks.

Execution Assumptions Market orders fill against simulated L2 depth. A fixed latency of 5 ms is applied, alongside a \$2.00 per-contract fee.

4.3 Backtest Design

Evaluated via NautilusTrader [6] using historical MBP-10 data. All tuning occurs strictly out-of-engine on in-sample splits.

5 Results

All figures in this section are hypothetical and illustrative. In-sample: 8 Jan–8 Feb 2024 (23 days; training and GA search). Out-of-sample: 8 Apr–8 Jul 2024 (63 days); the months between supply the purged walk-forward folds.

5.1 Performance Summary

Out-of-sample, the strategy takes 2,547 trades (40.4 per day). Before costs, 55.6% of trades win; winners average 4.68 ticks and losers 2.44 ticks, a gross edge of +1.52 ticks per trade. The 1.32-tick round-trip cost leaves +0.20 ticks (\$2.50) net, so costs absorb 87% of the gross edge. The break-even hit rate is $(2.44 + 1.32)/(4.68 + 2.44) \approx 52.8\%$, which the strategy clears by 2.8 points. The net Sharpe ratio falls from 2.78 in-sample to 1.58 out-of-sample, the expected cost of GA selection. The t -statistic over 63 days is 0.79, the probabilistic Sharpe ratio is 0.78, and the Deflated Sharpe Ratio [3], which accounts for about 400,000 GA trials, is 0.46. The level-1 baseline loses money, so the rule beats it.

Table 2: Performance statistics (net of costs)

Metric	In-Sample	Out-of-Sample	Benchmark
Annualized return (%)	42.3	25.5	−14.6
Annualized volatility (%)	15.2	16.1	17.3
Sharpe ratio	2.78	1.58	−0.84
Sortino ratio	4.42	2.39	−1.20
Max drawdown (%)	2.6	4.3	14.5
Calmar ratio	16.57	5.90	−1.01
Hit rate (%)	53.9	51.4	47.6
Turnover (% / month)	493,000	433,000	529,000

Benchmark: level-1 imbalance rule over the out-of-sample window. Returns use \$100k reference capital per contract and daily net P&L (252 days). Hit rate is per trade, net of costs. Turnover is traded notional ($\approx$ \$255k per contract side) over capital.

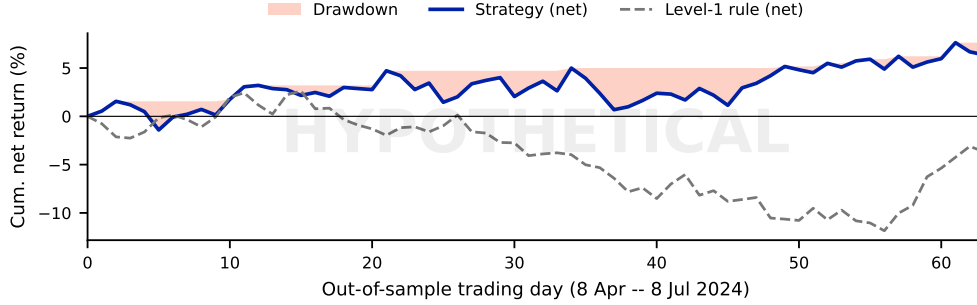


Figure 1: Cumulative net returns vs. benchmark.

5.2 Cumulative Performance

Figure 1 shows the hypothetical out-of-sample equity curve. The deepest drawdown is \$4,314 (4.3% of reference capital); three of four calendar months are profitable (Apr +\$2,075, May −\$440, Jun +\$3,454, 1–8 Jul +\$1,278).

5.3 Robustness and Attribution

Attribution. SHAP summaries (Appendix A, Figure 2) show that signal comes mainly from imbalance and activity features: `obi_10` ranks in the top two for both directional classes, `message_intensity` leads for long, and `deep_ask_slope` for flat. A bid-heavy top level pushes the model toward long and away from short, while `vpin` and `cancel_add_ratio` contribute almost nothing, so the GA can drop them, consistent with [2].

Cost and latency stress. Table 3 reruns the out-of-sample trades under alternative round-trip costs and fixed latencies. Profit survives up to about 1.5 ticks of cost or 10 ms of latency, but not 25 ms.

Table 3: Out-of-sample net P&L (\$) under cost and latency stress (hypothetical)

Round-trip cost (ticks)	1.00	1.32	1.50	2.00
Net P&L	16,556	6,367	637	−15,282
Latency (ms), cost 1.32	0	5	10	25
Net P&L	8,278	6,367	4,139	−1,592

Uncertainty. The bootstrap 95% interval for mean net P&L per trade is $[-\$3.7, +\$8.7]$ and for the annualized Sharpe ratio is $[-2.4, 5.5]$. Both include zero, so the sample is too short to separate

edge from noise.

6 Risk Management

6.0.1 Risk Controls

A rule-based risk manager overrides orders without changing size:

- **Pre-trade gates:** Blocks trades during maintenance breaks, major macro releases (FOMC, CPI), roll weeks, wide spreads, or extreme volatility regimes.
- **In-trade exits:** Enforces price stops, time stops (h), and pre-break flat triggers.
- **Session limits:** Halts trading on daily losses exceeding 2%, equity peak drawdowns of 5%, or sequential loss streaks.

7 Liquidity and Capacity

7.1 Liquidity Profile

ES trades about 1.4 million contracts per day, with a one-tick spread in nearly all book events and about 25 contracts at the top of book. Our 81 contracts per day (0.006% of ADV) fill at the first level.

7.2 Transaction Costs and Market Impact

Round-trip cost is 1.32 ticks (\$16.50: \$4.00 of fees plus one tick of spread), absorbing 87% of the 1.52-tick gross edge. Table 4 adds an uncalibrated square-root impact term of $0.008\sqrt{n}$ ticks per round trip for n contracts.

Table 4: Net Sharpe ratio vs. assumed AUM

AUM (\$M)	1	10	50	100
Net Sharpe	1.38	0.95	0.17	-0.42
Max ADV participation (%)	0.06	0.58	2.9	5.8

Hypothetical. One contract per \$100k of AUM (10, 100, 500, 1,000 contracts); 1.4M-contract ADV.

Net Sharpe falls below 1 near \$10M and turns negative by \$100M; edge is limited by cost, not liquidity.

8 Limitations and Next Steps

8.1 Limitations

- **Data:** trained on one month of ES only, which captures a single market regime; MBP-10 has no queue position.
- **Search budget:** compute limited the GA to 100 generations, so fitness may not have converged and results may be a local optimum.
- **Overfitting:** many parameter sets tested on overlapping returns from a short sample, so results may not generalize.
- **Regime risk:** untested under stress; bursts may not revert in high-volatility markets.
- **Execution:** assumes 1.32-tick round trips with no impact or latency model.

8.2 Next Steps

- **More data and generations:** ideally train on 2+ years, including 2022 and Aug. 2024, and run the GA until fitness plateaus; test on NQ.
- **Validation:** hold out a clean test period and fix overlapping fitness.
- **Execution:** simulate passive fills with MBO data, then paper-trade live in MES.

A SHAP Feature Attributions

Figure 2 shows per-class SHAP summaries of the XGBoost model (see Section 5).

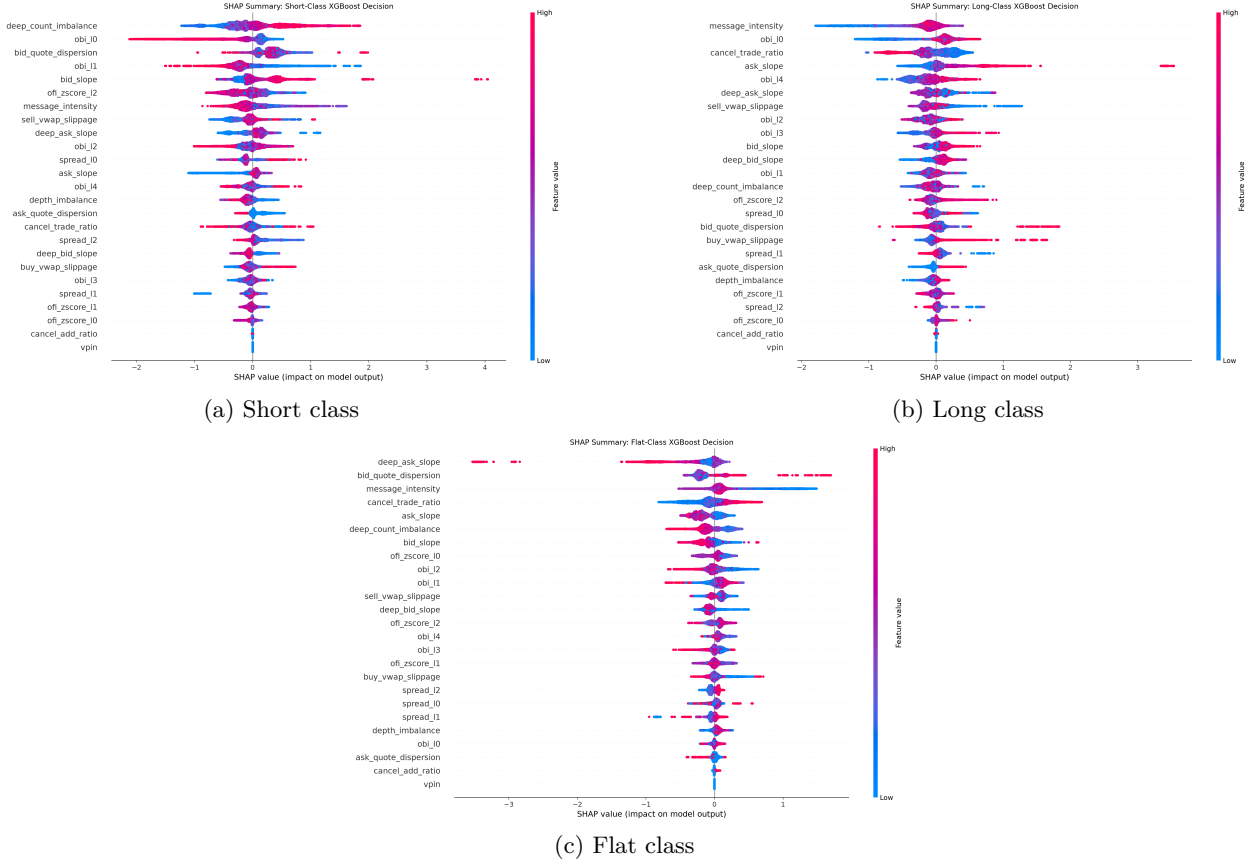


Figure 2: SHAP summary plots for the short-, long-, and flat-class XGBoost decisions.

References

- [1] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>
- [2] C. N. Christodoulou-Volos, “Optimizing Trading Strategies Using Genetic Algorithms: A Review and Implementation,” *Edelweiss Applied Science and Technology*, vol. 9, no. 10, pp. 334–345, 2025. <https://doi.org/10.55214/2576-8484.v9i10.10407>
- [3] D. H. Bailey and M. López de Prado, “The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality,” *Journal of Portfolio Management*, vol. 40, no. 5, pp. 94–107, 2014. <https://doi.org/10.3905/jpm.2014.40.5.094>
- [4] Databento. (2026). GLBX.MDP3: CME Globex market data, MBP-10 schema. Databento. <https://databento.com/catalog/cme/GLBX.MDP3>
- [5] P. Tillet, H. T. Kung, and D. Cox, “Triton: An Intermediate Language and Compiler for Tiled Neural Network Computations,” in *Proc. 3rd ACM SIGPLAN International Workshop on Machine Learning and Programming Languages (MAPL)*, 2019, pp. 10–19. <https://doi.org/10.1145/3315508.3329973>
- [6] Nautech Systems, “NautilusTrader,” software. https://github.com/nautechsystems/nautilus_trader